THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a circle of radius 5 cm, a chord is 3 cm from the centre. The length of the chord is: 1

(a) 10 cm **(b)** 6 cm **(c)** 8 cm **(d)** 4 cm

2. In a circle of radius 15 cm, a chord is 9 cm from the centre. The length of the chord is: 1

(a) 24 cm **(b)** 22 cm **(c)** 26 cm **(d)** 12 cm

3. A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. The radius of the circle is: 1

(a) 17 cm **(b)** 24 cm **(c)** 14 cm **(d)** 13 cm

4. A chord of length 14 cm is at a distance of 24 cm from the centre of a circle. The radius of the circle is: 1

(a) 31 cm **(b)** 25 cm **(c)** 26 cm **(d)** 14 cm

5. A chord of length 14 cm is drawn in a circle of radius 25 cm. The distance of the chord from the centre is: 1

(a) 8 cm **(b)** 18 cm **(c)** 32 cm **(d)** 24 cm

6. An arc subtends an angle of 84° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 138° **(b)** 84° **(c)** 168° **(d)** 42°

7. ABCD is a cyclic quadrilateral with $\angle A = 92^\circ$. Then $\angle C$ is: 1

(a) 44° **(b)** 92° **(c)** 88° **(d)** 268°

8. An arc subtends 41° at a point on the circle. The angle subtended by the same arc at the centre is: 1

(a) 41° **(b)** 139° **(c)** 42° **(d)** 82°

9. The greatest chord of a circle is: 1

(a) the diameter
(b) a chord at distance r from the centre
(c) the radius
(d) a chord of length equal to the radius

10. Of two chords of a circle, the one nearer to the centre is: 1

(a) cannot be compared **(b)** equal in length
(c) shorter **(d)** longer

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. AB is a diameter of a circle and C is a point on the circle. Find $\angle ACB$ and give the reason. 2

12. Show that the length of a chord at distance d from the centre of a circle of radius r is $2\sqrt{r^2 - d^2}$. 2

13. PQRS is cyclic with $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$. Find x. 2

14. ABCD is cyclic with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that equal chords of a circle are equidistant from the centre. 3

16. ABCD is a cyclic quadrilateral with $\angle ABC = 110^\circ$. Find the reflex angle AOC at the centre O, where the arc is ADC. 3

17. A triangle ABC is inscribed in a circle with centre O and $\angle BAC = 50^\circ$. Find $\angle BOC$ and $\angle OBC$. 3

OR

AB is a diameter of a circle, C is on the circle and $\angle CAB = 35^\circ$. Find $\angle ABC$. 3

18. If two equal chords of a circle intersect inside the circle, show that the segments of one chord are equal to the corresponding segments of the other. 3

OR

A cyclic quadrilateral has sides 5, 5, 12, 12 units in order. Find its area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Points A(0,0), B(8,0), C(0,6) and D(8,6) are corners of a rectangular plot. 5

(i) Find the length of diagonal BC.

(ii) Find the centre and radius of the circle through A, B, C.

(iii) Show that D lies on the same circle.

(iv) What type of quadrilateral is ABDC and why is it cyclic?

OR

Prove that if the opposite angles of a quadrilateral ABCD add up to 180°, then it is cyclic (do the case where C lies outside the circle through A, B, D). 5

20. Prove that the perpendicular bisector of a chord of a circle passes through the centre. 5

OR

ABCD is a cyclic quadrilateral with $\angle A = (3x + 5)^\circ$, $\angle B = (2y + 10)^\circ$, $\angle C = (5x - 25)^\circ$, $\angle D = (4y - 10)^\circ$. 5

(i) Find x and y.

(ii) Find all four angles.

(iii) Verify that the angle sum is 360°.

END OF QUESTION PAPER — Total: 40 Marks